Climate Sensitivity and Irrigation Buffering in Indian Rice Production:
Evidence from Maharashtra, Odisha, and Punjab (1990–2019)

Working Data Compilation

Kabir Suri

August 8, 2026

Note: This document compiles the data, equations, model output, and preliminary interpretation produced during this project's analysis phase. It is organized for drafting purposes, not a finished paper – prose, framing, and argument structure are left for the full write-up.

Contents

1 Data Summary

1.1 Sources

- **Yield and area data:** ICRISAT District Level Database (unapportioned), rice yield (kg/ha) and area sown (1000 ha), Maharashtra, Odisha ("Orissa" in source), and Punjab, 1990–2019.
- **Irrigation data:** ICRISAT District Level Database, cropwise irrigated area (1000 ha), rice.
- **Climate data:** India Meteorological Department (IMD) gridded daily rainfall ($0.25°$ resolution) and gridded daily maximum/minimum temperature ($1°$ resolution), downloaded via the `imdlib` Python package.
- **District boundaries:** Census 2011 district shapefile, used for spatial aggregation of climate grid cells to district level.

1.2 Cleaning and Sample Construction

- Raw yield file: 2,640 rows (88 districts × 30 years).
- Values of -1 treated as a missing-data placeholder (district not yet in existence / not reported); values of 0 in the yield file also treated as missing (implausible for an actively rice-growing district).
- Districts retained only if they had ≥ 20 valid years of yield data out of 30. This threshold, chosen *before* inspecting results, dropped 14 districts:
  Mumbai City, Mumbai Suburban, Palghar, Fazilka, Pathankot, Akola, Washim, Hingoli, Buldhana, Barnala, S.A.S. Nagar, Taran Taran, Jalna, Wardha.
- Most exclusions are explained by documented district boundary changes (Odisha's 1993 district reorganization; several Punjab and Maharashtra districts created post-1990) or negligible rice cultivation (Mumbai).
- **Final panel:** 74 districts × 30 years = 2,064 district-year observations.
  - Odisha: 30 districts, 815 obs.
  - Maharashtra: 27 districts, 758 obs.
  - Punjab: 17 districts, 491 obs.

1.3 Feature Construction

- **Growing season:** kharif season, June–October.
- **Rainfall/temperature anomalies:** for each district, $z_{it} = (x_{it} - \bar{x}_i)/s_i$, where $\bar{x}_i$ and s_i are the district's own baseline mean and standard deviation (1990–2010 baseline window).
- **Extreme heat days:** count of days with $T_{max} > 35°\text{C}$ during the growing season.
- **Max dry spell:** longest run of consecutive days with rainfall < 2.5mm during the growing season.

- **Irrigation share:** irrigated area/area sown, both from ICRISAT. Values > 1.0 (180 rows, nearly all ≤ 1.01 rounding artifacts in near-fully-irrigated Punjab districts, plus 3 genuine outliers in Ferozpur, Roopnagar, and Shri Mukatsar Sahib reaching 1.14–1.67) were capped at 1.0.

- **Irrigation data availability:** Maharashtra has no irrigation records in ICRISAT; `irrigation_share` is filled with 0 for all Maharashtra district-years. This is a *data-availability* zero, not a confirmed measured zero, and is **confounded with state identity** for Maharashtra – a key caveat for interpreting any irrigation-related coefficient or feature importance involving that state.

- **Lagged yield:** previous year's yield, same district. Undefined for each district's first sample year (1990), so those 74 rows are dropped in all regression/ML models. Final modeling sample: **1,990 observations**, 1991–2019.

1.4 Data Quality Checks Performed

- No duplicate district-year rows in the final merged panel.

- Merge diagnostics (indicator merges) confirmed 100% match rate between yield, climate, and irrigation sources on district–year keys.

- Value ranges checked as plausible: yield 79–5,077 kg/ha; kharif rainfall 62–5,687 mm; mean growing-season temperature 24.4–30.4°C.

- Anomaly variables confirmed properly standardized: mean ≈ 0, SD ≈ 1 across the full panel.

2 Methodology

2.1 Baseline Econometric Model

Two OLS specifications were estimated, with standard errors clustered by district throughout.

Model 1 (no fixed effects):

$$\begin{aligned}\ln(Yield_{it}) = \beta_0 + \beta_1\, RainfallAnomaly_{it} + \beta_2\, TempAnomaly_{it} + \beta_3\, ExtremeHeatAnomaly_{it} \\ + \beta_4\, DrySpellAnomaly_{it} + \beta_5\, IrrigationShare_{it} + \beta_6 \ln(Yield_{i,t-1}) + \varepsilon_{it}\end{aligned} \tag{1}$$

Model 2 (district and year fixed effects, main specification):

$$\begin{aligned}\ln(Yield_{it}) = \beta_1\, RainfallAnomaly_{it} + \beta_2\, TempAnomaly_{it} + \beta_3\, ExtremeHeatAnomaly_{it} \\ + \beta_4\, DrySpellAnomaly_{it} + \beta_5\, IrrigationShare_{it} + \beta_6 \ln(Yield_{i,t-1}) + \delta_i + \gamma_t + \varepsilon_{it}\end{aligned} \tag{2}$$

where i indexes district, t indexes year, δ_i is a district fixed effect (absorbing time-invariant district characteristics such as soil quality), and γ_t is a year fixed effect (absorbing nationwide shocks common to all districts in a given year). Since the dependent variable is logged, each $\hat{\beta}$ is approximately interpretable as a percentage change in yield per one-unit (here, one-standard-deviation, since anomaly variables are z-scores) increase in the corresponding regressor.

2.2 Machine Learning Model

A `RandomForestRegressor` (scikit-learn; 500 trees, `min_samples_leaf`=2) was fit on the same six predictors (using ln(lagged yield) in place of the raw level, for scale consistency with the target). Two validation strategies were used:

1. **Temporal holdout:** trained on 1991–2009 (1,281 obs.), tested on 2010–2019 (709 obs.) – a genuine forecasting test, since a random split would leak information across adjacent years within the same districts.
2. **District-grouped cross-validation:** 5-fold `GroupKFold`, grouped by district (not random k-fold, since observations from the same district over time are not independent).

2.3 District Climate Sensitivity Simulation

For each district i, using only its 1991–2009 historical average feature values $\bar{x}_i$ (all features except rainfall anomaly held fixed):

$$\hat{Y}_i^{normal} = \exp\left(f_{RF}(\bar{x}_i \mid RainfallAnomaly = 0)\right) \tag{3}$$

$$\hat{Y}_i^{shock} = \exp\left(f_{RF}(\bar{x}_i \mid RainfallAnomaly = -1)\right) \tag{4}$$

$$\text{PredictedLoss}_i\,(\%) = \frac{\hat{Y}_i^{normal} - \hat{Y}_i^{shock}}{\hat{Y}_i^{normal}} \times 100 \tag{5}$$

where $f_{RF}(\cdot)$ is the fitted random forest's log-yield prediction function. This produces a *model-based predictive sensitivity ranking*, not a causal estimate of rainfall's effect on yield.

3 Results

3.1 Baseline Regression Results

Table 1: OLS Regression Results (Dependent variable: ln Yield)

	Model 1 (No FE)	Model 2 (District + Year FE)
Rainfall anomaly	0.069^{***} (0.014)	0.030^{***} (0.010)
Temperature anomaly	-0.021^{**} (0.009)	-0.034^{***} (0.011)
Extreme heat anomaly	0.016 (0.010)	0.021^{*} (0.013)
Dry spell anomaly	-0.017^{**} (0.008)	-0.027^{***} (0.009)
Irrigation share	0.392^{***} (0.075)	0.417^{***} (0.155)
ln(Lagged yield)	0.687^{***} (0.052)	0.234^{***} (0.084)
Constant	2.157^{***} (0.358)	– (absorbed by δ_i)
District FE	No	Yes (74)
Year FE	No	Yes (30)
R^2	0.702	0.817
N	1,990	1,990

Clustered SE (district) in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Note: Model 2 covariance matrix showed reduced rank (34 of 107 constraints) due to district/year FE collinearity in short district panels; coefficient signs/significance were stable across specifications.

Key observations:

- All climate variables carry theoretically expected signs and are statistically significant at conventional levels in Model 2, except extreme heat anomaly (significant only at 10%).
- Irrigation share is the single largest and most robust coefficient: full irrigation ($0 \to 1$) is associated with approximately 42% higher yield, holding climate and lagged yield fixed.
- The lagged-yield coefficient falls sharply between Model 1 and Model 2 ($0.687 \to 0.234$). This indicates that a substantial share of the apparent yield persistence in the naive model reflects fixed district characteristics (e.g., soil quality) rather than genuine year-to-year dynamic persistence.

3.2 Random Forest Performance

Feature importances: ln(lagged yield) 0.722, irrigation share 0.115, rainfall anomaly 0.067, extreme heat anomaly 0.034, dry spell anomaly 0.032, temperature anomaly 0.030. Feature importance measures predictive contribution within the fitted model, not a causal effect.

3.2.1 Weak Cross-Validation Fold

The lowest-scoring fold ($R^2 = 0.316$) contained 15 districts, exclusively from Maharashtra (9) and Odisha (6) – no Punjab districts. Relative to the remaining 59 districts, this group had substantially lower mean irrigation share (0.143 vs. 0.446) and lower mean yield (1,130 vs. 2,016 kg/ha). This indicates the model's predictive accuracy is systematically weaker for low-irrigation,

Table 2: Model comparison, out-of-sample (2010–2019 test period)

Model	Test R^2 (kg/ha scale)	Test RMSE (kg/ha)
Random forest	**0.754**	**614.0**
Naive (yield = lagged yield)	0.716	660.3
OLS (district + year FE)†	0.691	688.0

†Refit on 1991–2009 only; year fixed effects for 2010–2019 unobserved and proxied by the average training-period year effect. Not directly comparable to the full-sample Model 2 R^2 of 0.817 reported above, which is in-sample.

Table 3: Random forest fit and validation summary

Metric	Value
Train R^2 (1991–2009), log scale	0.955
Test R^2 (2010–2019), log scale	0.669
5-fold GroupKFold mean R^2 (by district)	0.719
5-fold GroupKFold R^2 SD	0.206
GroupKFold fold scores	0.763, 0.821, 0.807, 0.316, 0.889

rainfed production systems – precisely the districts most relevant to the project's climate-risk policy question.

3.3 District Climate Sensitivity Ranking

Table 4: State-level average predicted yield loss under a −1SD rainfall shock

State	Avg. loss (%)	Avg. irrigation share	Districts	Min loss (%)	Max loss (%)
Maharashtra	19.49	0.000	27	0.69	35.72
Odisha (Orissa)	19.26	0.400	30	0.53	35.41
Punjab	−0.45	0.990	17	−1.28	0.40

Table 5: Top 10 most climate-sensitive districts

District	State	Irrigation share	Predicted loss (%)	Weak CV fold?
Gondia	Maharashtra	0.000	35.72	Yes
Nawarangpur	Odisha	0.071	35.41	Yes
Dhenkanal	Odisha	0.285	34.78	No
Jharsuguda	Odisha	0.159	34.64	Yes
Nasik	Maharashtra	0.000	34.58	Yes
Phulbani (Kandhamal)	Odisha	0.243	32.71	No
Mayurbhanja	Odisha	0.240	32.69	No
Nayagarh	Odisha	0.273	31.99	No
Bolangir	Odisha	0.212	31.50	No
Keonjhar	Odisha	0.249	31.30	No

Table 6: Bottom 10 (least sensitive) districts, all Punjab

District	Irrigation share	Predicted loss (%)
Gurdaspur	0.932	−0.59
Shri Mukatsar Sahib	0.997	−0.95
Hoshiarpur	0.981	−0.96
Amritsar	0.998	−0.99
Ferozpur	0.995	−1.07
Faridkot	0.998	−1.17
Mansa	0.999	−1.28

4 Preliminary Economic Interpretation

4.1 Irrigation as a Climate Buffer – and Its Nonlinearity

Both the fixed-effects regression and the random forest simulation independently support the same core conclusion: irrigation coverage substantially buffers rice yield against rainfall variability. The regression estimates a ∼42% yield gain from full irrigation; the simulation shows near-zero predicted rainfall-shock sensitivity specifically in Punjab, where irrigation coverage approaches 100%.

The pattern is, however, **not linear in irrigation share**. Odisha's district-average irrigation share (0.40) is meaningfully higher than Maharashtra's (0.00, though partly a data-availability artifact), yet the two states show nearly identical average climate sensitivity (∼19.3–19.5%). This suggests a **threshold effect**: partial irrigation coverage in the 20–50% range does not translate into proportionally lower climate risk, and only near-complete coverage (Punjab's regime) delivers a strong protective effect. A plausible economic mechanism: partial irrigation may cover only a portion of a district's rice area or provide supplemental (not primary) water in dry years, leaving yield still substantially rainfall-dependent unless coverage is comprehensive.

4.2 Policy Implication

If the threshold interpretation holds, incremental irrigation investment in the 0–50% coverage range may yield limited climate-risk reduction, while investment aimed at pushing coverage toward near-universal levels (as in Punjab) could deliver disproportionately large risk reduction. This has direct relevance for targeting decisions under schemes such as PMFBY (crop insurance premium differentiation) or state irrigation investment planning – though this inference rests on cross-sectional variation across just three states and should be treated as suggestive rather than confirmed.

4.3 A Note on the Model's Own Blind Spot

The district-grouped cross-validation result implies a genuine tension: the model is least reliable precisely for the low-irrigation, rainfed districts that a climate-risk targeting exercise would most want to prioritize. Any policy application of this ranking should weight the qualitative rainfed/irrigated distinction alongside the specific numeric ranking, rather than treating the point estimates as equally precise across all districts.

5 Known Limitations (for full writeup)

- **Irrigation–state confounding:** Maharashtra's `irrigation_share` is a filled zero (data unavailable), not a measured value, and is perfectly collinear with Maharashtra state identity. Any irrigation coefficient or importance score is partly a state effect.
- **Heterogeneous model reliability:** cross-validated R^2 ranges from 0.32 to 0.89 across district groups; the weakest group is disproportionately rainfed and low-yield.
- **Reduced-rank covariance warning** in Model 2's fixed-effects specification (statsmodels), due to collinearity among many district/year dummies in a moderately short panel; coefficient signs and significance were stable, but exact standard error precision should be treated with mild caution.
- **District panel imbalance from boundary changes:** 14 of 88 original districts excluded due to insufficient valid-year coverage, mostly from post-1990 district creation (Odisha's 1993 reorganization; various Punjab/Maharashtra splits).
- **Small number of measurement outliers** in the irrigation/area merge (Ferozpur 2012/2013/2019; Roopnagar 1992; Shri Mukatsar Sahib 1997), where reported irrigated area exceeded reported area sown by 14–67%, capped at 1.0 rather than treated as reliable values.
- **Feature importance $\neq$ causal effect** throughout; the risk ranking is a model-based predictive simulation, not an estimated causal treatment effect of rainfall.
- **Moderate sample size** for ML methods: 74 districts, 30 years – adequate for the methods used, but modest relative to typical ML applications, contributing to the cross-validation variance observed.